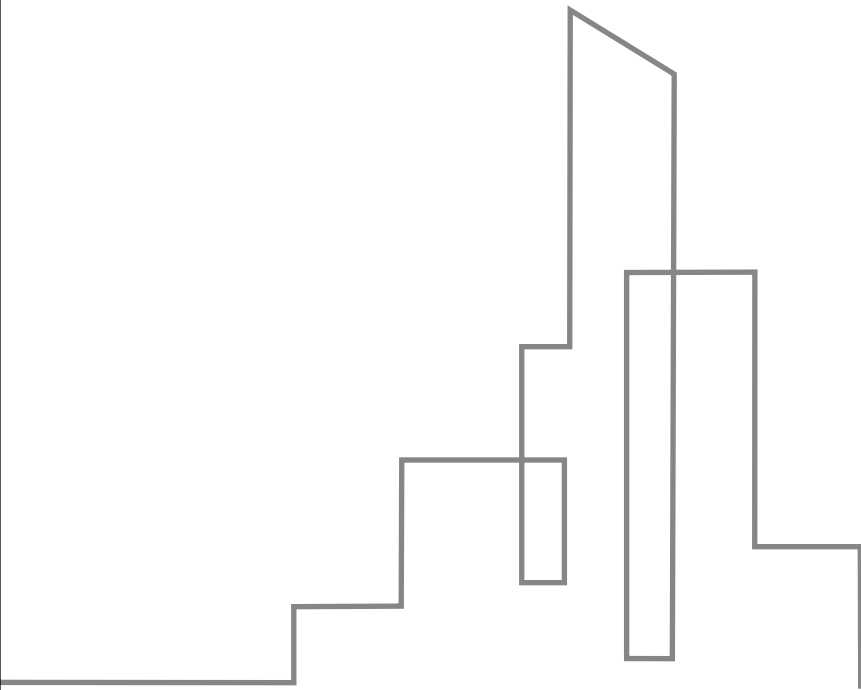


PE and CE Integration



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CE Components in
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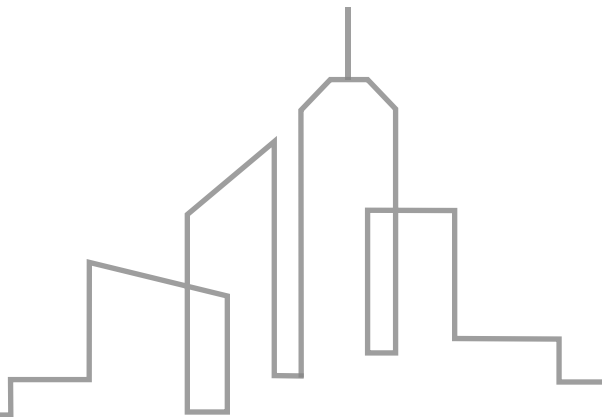
04

Conclusion: Combined
Approach



01

Introduction: PE vs. CE





PE: Crafting static instructions

Role Assignment

Defining the Agent's professional persona, e.g., Senior Security Analyst.

Few-Shot Examples

Showing input/output pairs for consistent formatting, e.g., JSON.

Chain of Thought (CoT)

Forcing step-by-step reasoning to ensure logical consistency.

Constraint Setting

Defining explicit boundaries for the analysis to maintain focus.



CE: Assembling dynamic resources

01

Memory & State Management

Short-Term Memory for conversation summaries and Long-Term Memory for vector databases.

02

RAG (Retrieval Augmented Generation)

Connecting the Agent to proprietary, dynamic data using hybrid search.

03

Tools for Real-World Actions

External interfaces like APIs, SQL, and Code Execution enabling real-world tasks.



CE necessity for SaaS

Reliable Data-Intensive Features

CE is crucial for creating systems that handle large volumes of data reliably.

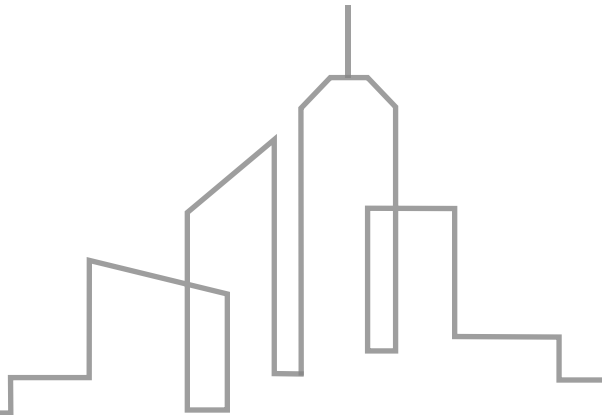


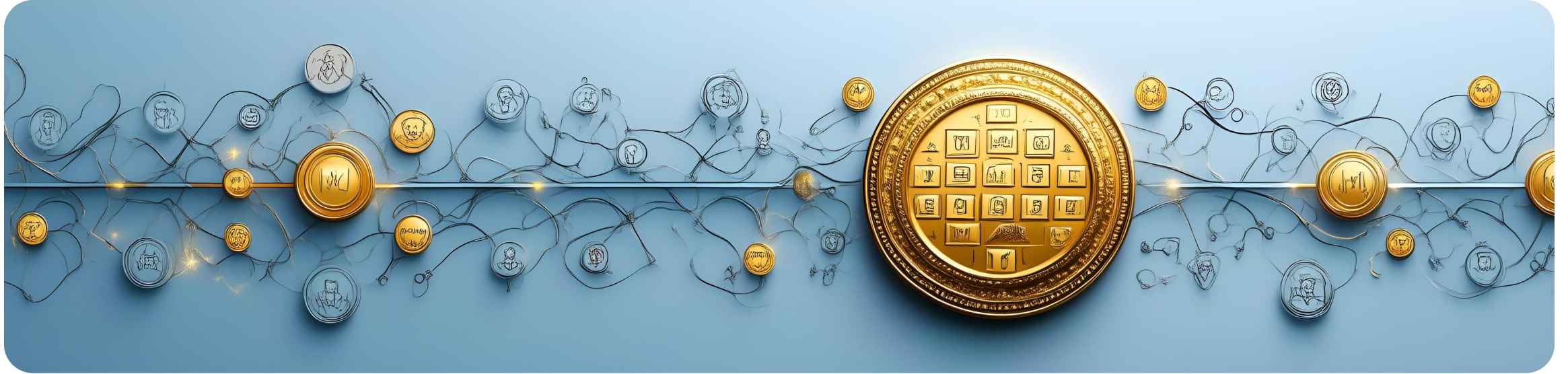
Transition to Autonomous Agent

Integrating PE and CE content to evolve the system into an autonomous agent.

02

Core PE Techniques





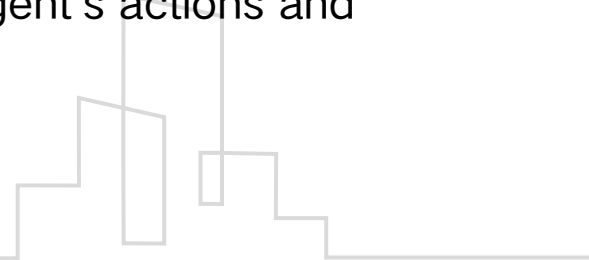
Role Assignment for Agent

Defining Agent's Persona

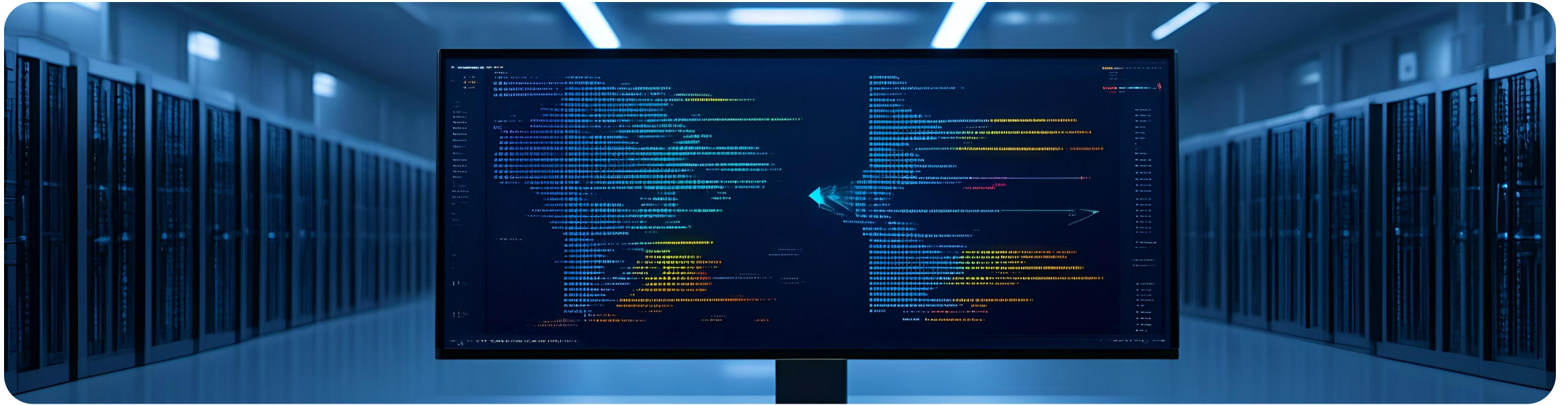
Assigning a professional role such as Senior Security Analyst to the Agent.

Guiding Agent's Behavior

Role assignment steers the Agent's actions and decision-making process.



Few-Shot Examples for Formatting



Showing Input/Output Pairs

Providing examples like JSON to ensure consistent formatting.

Consistency in Formatting

Few-shot examples help maintain uniformity in data presentation.

Chain of Thought for Reasoning



Step-by-Step Reasoning

Forcing the Agent to explain its reasoning process in a logical sequence.



Enhancing Comprehension

CoT improves understanding of the Agent's decision-making pathway.

Constraint Setting for Analysis

Defining Explicit Boundaries

Setting constraints to limit the scope of the Agent's analysis.

Guiding Accurate Analysis

Explicit constraints ensure the Agent's analysis remains focused and precise.

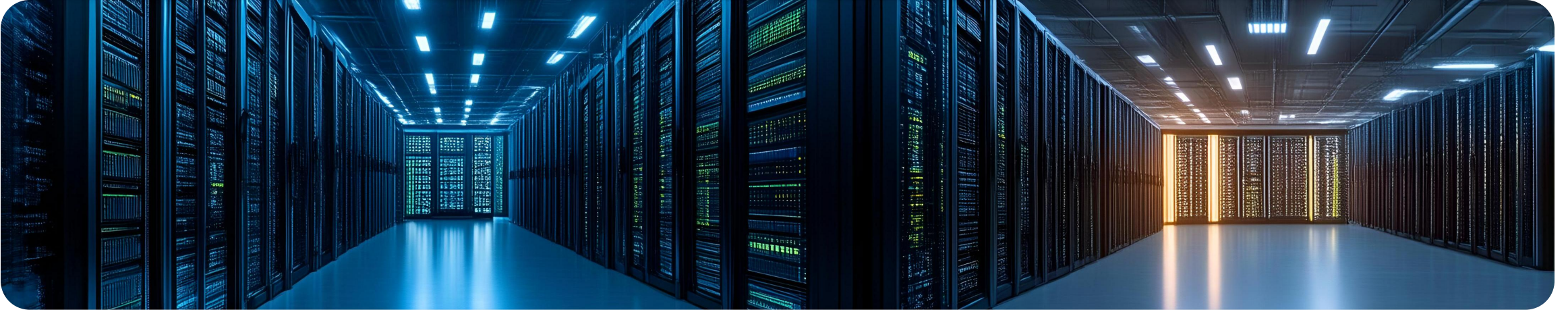


03

CE Components in Agentic Systems



Memory & State Management



Short-Term Memory

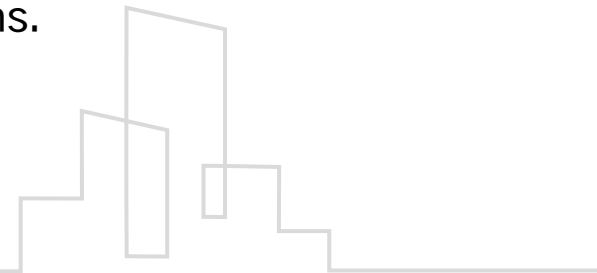
Holds conversation summaries for immediate context retention.

Long-Term Memory

Utilizes vector databases to store learned patterns for future reference.

State Management

Tracks the status of multi-step processes to maintain coherent interactions.



RAG for Dynamic Data Connection



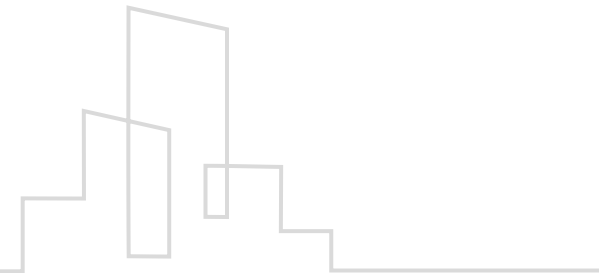
Proprietary Data Connection

RAG links the Agent to client-specific, dynamic data.



Hybrid Search Utilization

Employing hybrid search to retrieve contextually relevant document sections.

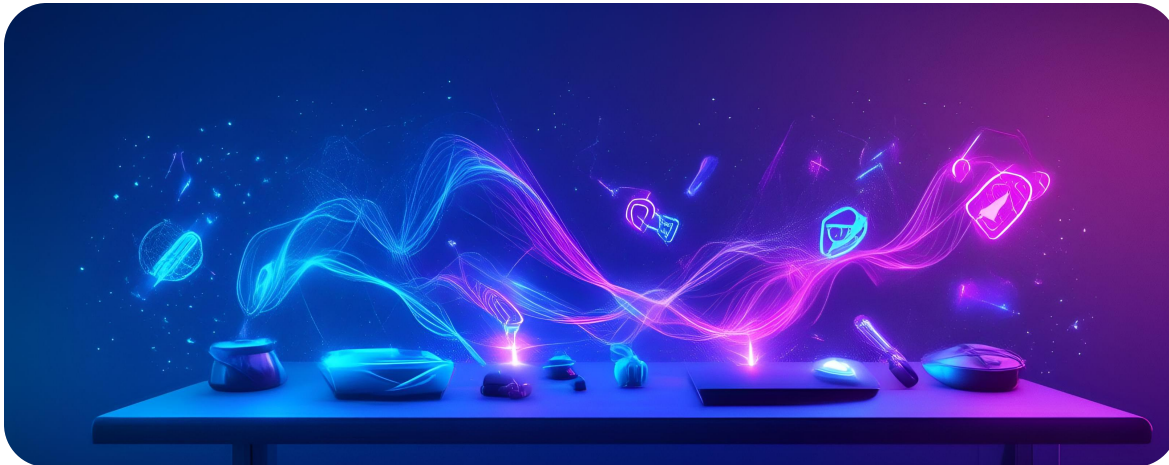


Tools for Real-World Actions



External Interfaces

APIs, SQL, and Code Execution enable real-world actions.



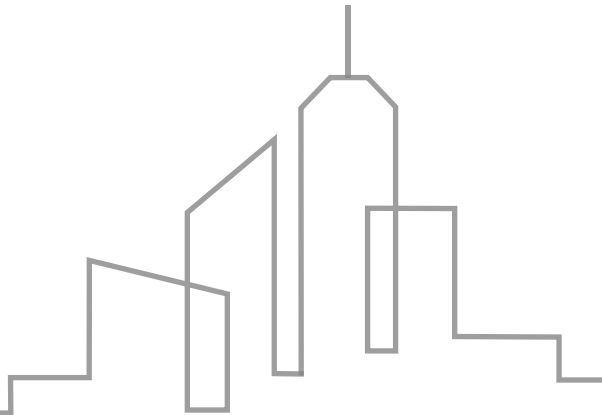
CE Directs LLM Usage

Context Engineering defines the application of tools by the LLM.



04

Conclusion: Combined Approach



Blending PE and CE

Static and Dynamic Integration

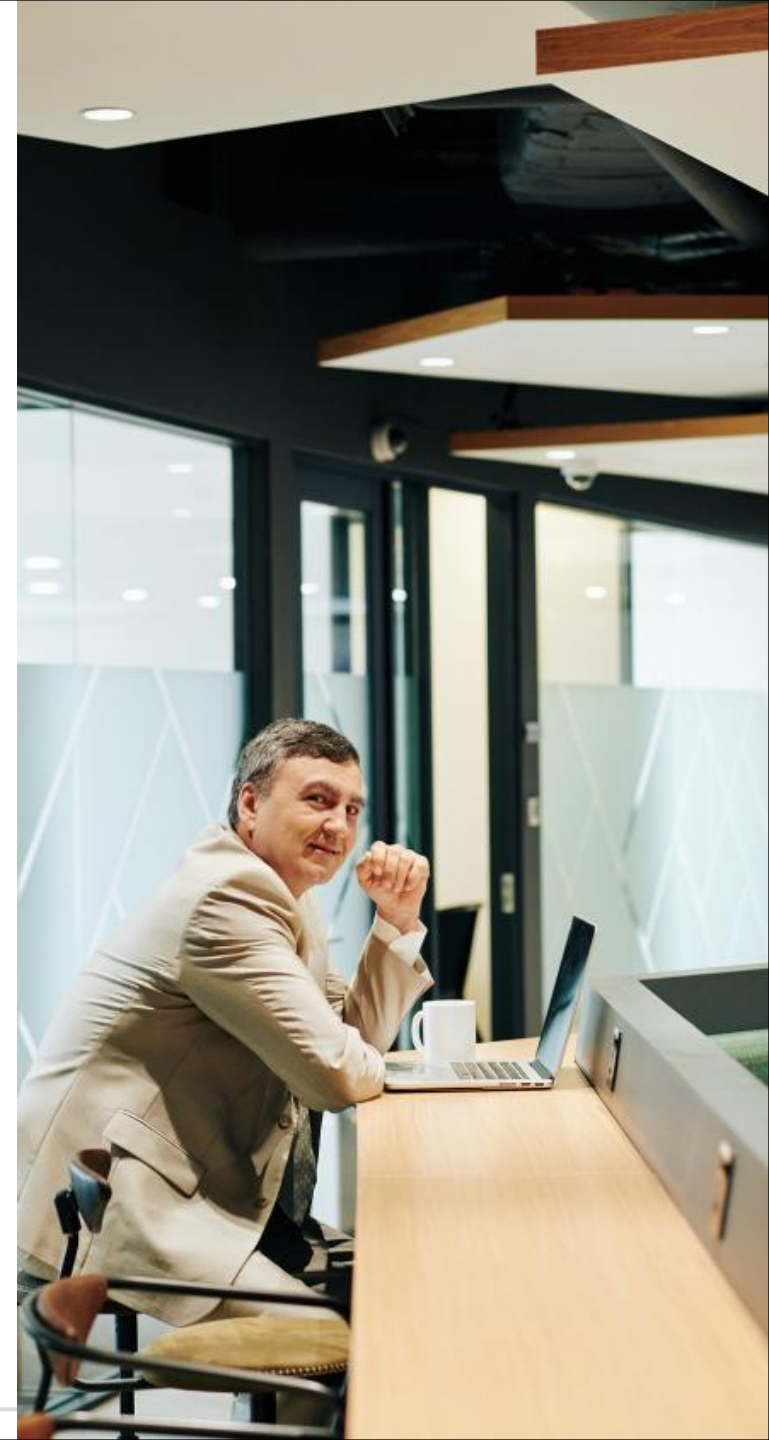
The final prompt combines PE instructions with dynamic CE content for a comprehensive approach.

RAG and Memory Utilization

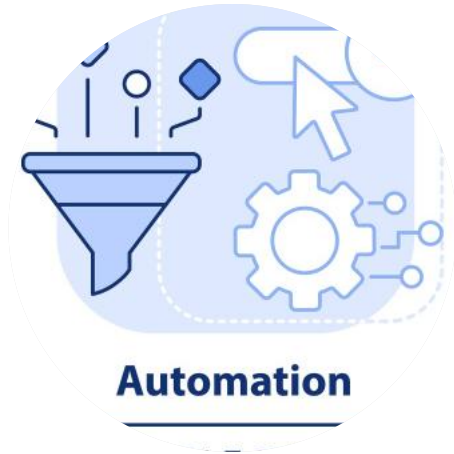
RAG connects to dynamic data, while Memory & State Management tracks the process status.

Transition to Autonomous Agent

The integration of PE and CE components is key to evolving the system into an autonomous agent.



Transition to Autonomous Agent



Evolution from Simple AI

The combined approach enables the transition from basic AI to an autonomous, reliable agent.



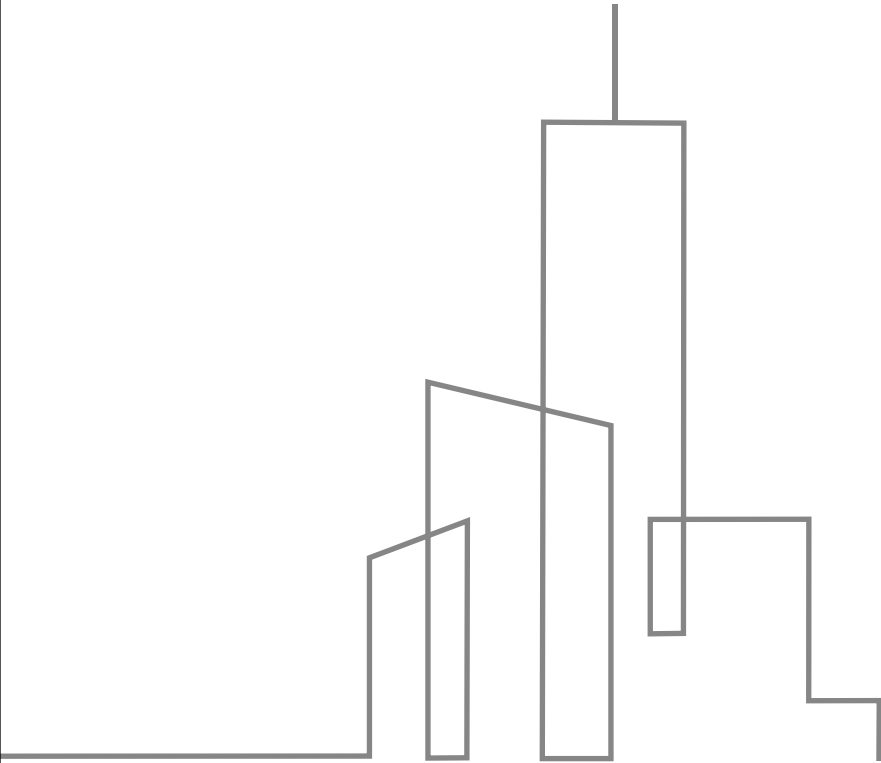
Reliable saascv Agent

CE's role in data-intensive features ensures the agent's reliability in saascv tasks.



Autonomous Capabilities

The autonomous agent can perform real-world actions through external tools and APIs defined by CE.



THE END

Thank You